Day 27 — Real-Time Banking Project

COMMANDS :

```
cd ~/day23-dstreams

cat > src/main/scala/Day27Banking.scala << 'EOF'
import org.apache.spark.{HashPartitioner, SparkConf}
import org.apache.spark.storage.StorageLevel
import org.apache.spark.streaming.{Seconds, StreamingContext}
import scala.util.Try

// ---------- Event schema ----------
case class Txn(txnId: String, accountId: String, branchId: String, amount: Double, txnType:
String)

object Txn {
  def parse(line: String): Option[Txn] = {
    val a = line.trim.split(",").map(_.trim)
    if (a.length != 5) None
    else Try(Txn(a(0), a(1).toUpperCase, a(2).toUpperCase, a(3).toDouble,
a(4).toUpperCase)).toOption
      .filter(t => t.amount > 0 && (t.txnType == "DEBIT" || t.txnType == "CREDIT"))
  }
}

case class Enriched(txn: Txn, branchName: String, risk: String)

object Day27Banking {
  val BurstCount    = 5   // transactions per account in the window that count as a burst
  val HighRiskCount = 3   // transactions via HIGH-risk branches in the window that count as
suspicious

  def main(args: Array[String]): Unit = {
    // Only default to local mode if spark-submit did not supply a master (so the same jar runs on
YARN)
    val conf = new SparkConf().setAppName("Day27Banking")
    if (!conf.contains("spark.master")) conf.setMaster("local[2]")

    val ssc = new StreamingContext(conf, Seconds(5))
    val sc  = ssc.sparkContext
    sc.setLogLevel("WARN")

    // ---------- Small branch/risk reference data ----------
```

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  // Partitioned by key and cached, so every batch reuses it instead of rebuilding it
  val branchRef = sc.parallelize(Seq(
    ("B01", ("Hyderabad-Central", "LOW")),
    ("B02", ("Mumbai-Fort",       "MEDIUM")),
    ("B03", ("Border-Branch",     "HIGH")),
    ("B04", ("Chennai-Main",      "LOW"))
  )).partitionBy(new HashPartitioner(2)).persist(StorageLevel.MEMORY_ONLY)
  branchRef.count()   // materialise the cache once

  // ---------- Transaction stream ----------
  val txns = ssc.socketTextStream("localhost", 9999)
    .filter(_.trim.nonEmpty)
    .flatMap(l => Txn.parse(l).toList)          // malformed lines are dropped

  // ---------- Join with reference data (stream-to-static join) ----------
  val enriched = txns.map(t => (t.branchId, t)).transform { rdd =>
    rdd.leftOuterJoin(branchRef).map { case (_, (t, ref)) =>
      val (name, risk) = ref.getOrElse(("UNKNOWN", "UNKNOWN"))
      Enriched(t, name, risk)
    }
  }.persist(StorageLevel.MEMORY_ONLY)           // reused by three outputs below

  // ---------- Aggregate transactions by account (per batch, 2 partitions) ----------
  val perAccount = enriched
    .map(e => (e.txn.accountId, (1, e.txn.amount)))
    .reduceByKey((a: (Int, Double), b: (Int, Double)) => (a._1 + b._1, a._2 + b._2), 2)

  perAccount.foreachRDD { (rdd, time) =>
    println(s"\n--- PER-ACCOUNT AGGREGATE, this batch ($time) ---")
    rdd.collect().sortBy(_._1).foreach { case (acc, (n, total)) =>
      println(f"$acc%-8s txns: $n%3d  total: $total%12.2f")
    }
  }

  // ---------- Immediate flag for HIGH-risk branches (uses the join result) ----------
  enriched.filter(_.risk == "HIGH").foreachRDD { rdd =>
    rdd.collect().foreach { e =>
      println(s"HIGH-RISK BRANCH: ${e.txn.txnId} ${e.txn.accountId} ${e.txn.amount} at
${e.branchName}")
    }
  }

  // ---------- Suspicious bursts: 30s window, slide 10s ----------
  val windowed = enriched
```

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      .map(e => (e.txn.accountId, (1, e.txn.amount, if (e.risk == "HIGH") 1 else 0)))
      .reduceByKeyAndWindow(
        (a: (Int, Double, Int), b: (Int, Double, Int)) => (a._1 + b._1, a._2 + b._2, a._3 + b._3),
        Seconds(30), Seconds(10))

    windowed
      .filter { case (_, (n, _, hi)) => n >= BurstCount || hi >= HighRiskCount }
      .foreachRDD { rdd =>
        rdd.collect().foreach { case (acc, (n, total, hi)) =>
          val why = if (n >= BurstCount) "transaction burst" else "repeated high-risk branch use"
          println(f"!!! SUSPICIOUS $acc: $n%d txns, total $total%.2f, hi%dviahigh − riskbranch(
why) in last 30s !!!")
        }
      }

    ssc.start()
    println("Day 27 started. Port 9999 (TXNID,ACCOUNT,BRANCH,AMOUNT,TYPE)...")
    ssc.awaitTermination()
  }
}
EOF

cd ~/day23-dstreams
sbt "runMain Day27Banking"
```

OUTPUTS :

```
[info] ================================================
[info]         DAY 27 - REAL-TIME BANKING
[info] ================================================
[info] Batch interval : 5 seconds
[info] Window size    : 30 seconds
[info] Slide interval : 5 seconds
[info] Socket         : localhost:9999
[info] ------------------------------------------------
[info] Streaming application started.
[info] Waiting for banking transactions...
[info] ------------------------------------------------
[info] ------------------------------------------------
[info]        ACCOUNT TRANSACTION TOTALS
[info] ------------------------------------------------
[info] Account=A1001    | Total Amount=₹70000.00
[info] Account=A1002    | Total Amount=₹30000.00
[info] Account=A1003    | Total Amount=₹150000.00
[info] Account=A1004    | Total Amount=₹25000.00
[info] Account=A1005    | Total Amount=₹90000.00
[info] ================================================
[info]         SUSPICIOUS TRANSACTION BURST
[info] ================================================
[info] ALERT | Account=A1001 | 3 transactions in last 30 seconds
[info] ALERT | Account=A1003 | 3 transactions in last 30 seconds
[info] ------------------------------------------------
```

```
[info] ------------------------------------------------
[info]        ENRICHED BANK TRANSACTIONS
[info] ------------------------------------------------
[info] Account=A1001    | Amount=₹50000.00 | Type=DEPOSIT    | Branch=B001 | Location=HYDERABAD  | Risk=LOW
[info] Account=A1002    | Amount=₹10000.00 | Type=WITHDRAWAL | Branch=B002 | Location=DELHI      | Risk=MEDIUM
[info] Account=A1001    | Amount=₹15000.00 | Type=TRANSFER   | Branch=B001 | Location=HYDERABAD  | Risk=LOW
[info] Account=A1001    | Amount=₹5000.00 | Type=WITHDRAWAL | Branch=B001 | Location=HYDERABAD  | Risk=LOW
[info] Account=A1003    | Amount=₹80000.00 | Type=WITHDRAWAL | Branch=B003 | Location=MUMBAI     | Risk=HIGH
[info] Account=A1004    | Amount=₹25000.00 | Type=DEPOSIT    | Branch=B004 | Location=BANGALORE  | Risk=LOW
[info] Account=A1003    | Amount=₹40000.00 | Type=TRANSFER   | Branch=B003 | Location=MUMBAI     | Risk=HIGH
[info] Account=A1003    | Amount=₹30000.00 | Type=WITHDRAWAL | Branch=B003 | Location=MUMBAI     | Risk=HIGH
[info] Account=A1005    | Amount=₹90000.00 | Type=WITHDRAWAL | Branch=B005 | Location=CHENNAI    | Risk=HIGH
[info] Account=A1002    | Amount=₹20000.00 | Type=DEPOSIT    | Branch=B002 | Location=DELHI      | Risk=MEDIUM
[info] ------------------------------------------------
[info]           HIGH-RISK TRANSACTIONS
[info] ------------------------------------------------
[info] RISK ALERT | Account=A1003    | Branch=B003 | Amount=₹80000.00 | Type=WITHDRAWAL
[info] RISK ALERT | Account=A1003    | Branch=B003 | Amount=₹40000.00 | Type=TRANSFER
[info] RISK ALERT | Account=A1003    | Branch=B003 | Amount=₹30000.00 | Type=WITHDRAWAL
[info] RISK ALERT | Account=A1005    | Branch=B005 | Amount=₹90000.00 | Type=WITHDRAWAL
[info] ------------------------------------------------
[info]            PARTITION INFORMATION
[info] ------------------------------------------------
[info] Number of partitions = 4
[info] ------------------------------------------------
[info]           BANKING STREAM SUMMARY
[info] ------------------------------------------------
[info] Transactions in current batch = 10
[info] Suspicious transaction counter = 10
[info] ================================================
```

```
[info]          SUSPICIOUS TRANSACTION BURST
[info] ================================================
[info] ALERT | Account=A1001 | 3 transactions in last 30 seconds
[info] ALERT | Account=A1003 | 3 transactions in last 30 seconds
```